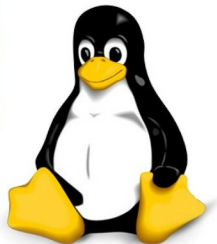
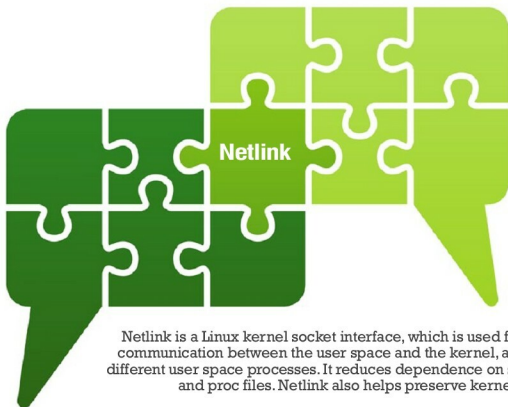


Netlink: A Communication Mechanism in Linux



Netlink is a Linux kernel socket interface, which is used for inter-process communication between the user space and the kernel, and also between different user space processes. It reduces dependence on system calls, `ioctl`s and `proc` files. Netlink also helps preserve kernel purity.

There are various ways by which processes in the user space can communicate with the kernel. These are system calls, `ioctl` and the `proc` file system. The problem with system calls is that they are linked statically to kernel code. So, any new feature that is to be provided has to be compiled with the kernel, but with dynamic linking modules such as device drivers, any feature that these modules want to provide cannot always be preconfigured as a system call. Similarly, for every new feature, it is hard to provide communication through the file system. All these communication mechanisms also require that the processes initiate the communication – the kernel cannot initiate the communication.

Netlink socket

Netlink socket is a communication mechanism used between the user space processes and also for communication between processes and the kernel. It can also be used for communication between user space threads and the kernel. It is a full duplex communication mechanism, that is, the kernel itself can initiate the communication. One of the advantages of this mechanism is that, in the user space, popular socket APIs that software programmers are familiar with are used for Netlink communication, so no new study is required. Netlink sockets are easier to add than system calls, `ioctl`s and `proc` files—all of which will start polluting

the kernel. If these are added for every new feature, then later on, it becomes difficult to remove these features -- a problem kernel developers are facing currently in the case of the `proc` file system. In case of Netlink sockets, only a protocol type macro needs to be inserted in the `netlink.h` file, which resides in `include/uapi/linux/netlink.h`, and processes and the kernel can start communicating immediately through the socket API. Netlink socket is an asynchronous communication method, that is, it queues the messages to be sent in the receiver's Netlink queue. One of the features of a Netlink socket is that it also supports multicast communication, i.e., one process can send a message to a Netlink group address, and many processes can listen on this group address. Since in the user space it is implemented through the socket API, this is an easy-to-use communication mechanism.

The basics of Netlink sockets

To use Netlink sockets in code, a standard socket API is used, which is as follows:

```
int socket(int domain, int type, int protocol);
```

Here, `domain` specifies the protocol family used for communication which is defined in `sys/socket.h`; `domain` in the case of Netlink is `AF_NETLINK`.